

EE 446: TinyML

Lab 9: American Sign Language Recognition

Dept. of Electrical and Computer Engineering
University of Washington

Instructor: Dinuka Sahabandu (sdinuka@uw.edu)



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Today's Lab

- Both Software and Hardware Lab!
- Training models to recognize the alphabet
- Quantize the model for TinyML
- Deploy the model to the Arduino board



Lab 9 - Software - Road Map

1. Gesture Based ASL recognition
2. Model training
3. Model Evaluation
4. Covert the TensorFlow model to TensorFlow Lite Micro model
5. Compress the model



Lab 9 - Hardware - Road Map

1. Connect Arduino Boards to your computer
2. Implement ASL recognition using compressed ML model
3. Deploy the model on to board
4. Open Serial Monitor and test the model



Training and Quantize the Model

Training and Quantizing the Model

1. Open the notebook

Open: `EE446_TinyML_Lab8.ipynb` in **JupyterLab** using the `tinymml-arduino` environment.

2. Build, train, and quantize the model

Train the gesture classification model and convert it to a TensorFlow Lite model.

3. Export the model for Arduino deployment

Export the trained model as a C/C++ header file: `model.h`

This file contains the TFLite model as a byte array that can be included in the Arduino sketch (.ino file).



TinyML Detective Challenge

- We intentionally introduced **one issue** in the **Build and Train the Model** section.
- **Find and correct the issue before training the model.**
- After fixing it, make sure any downstream **plots, axis labels, metrics, history keys, and titles** remain consistent with your correction.



Deploy the Model

- Connect the **Arduino Nano 33 BLE Sense or Rev2** to your computer using a USB cable. You should see the green power LED turn on.
- Open the Arduino sketch provided under the **Hardware** folder: lab8-classifier.ino
- The provided .ino file is written to support both board types: **Nano 33 BLE Sense Rev2** and **original Nano 33 BLE Sense / TinyML Kit board**. Depending on the board you are using, adjust the board setting near the top of the sketch before uploading.
- In the Arduino IDE, select the correct:
 - **Tools → Board**
 - **Tools → Port**
- Click the **Upload** button to flash the code to the board.
- Open the **Serial Monitor** and test your gesture classifier.
- The Serial Monitor will print the predicted gesture and confidence score. Move the board, stop, and wait for the model to process the gesture.



Submission Instructions

- Complete **both parts** of this lab:

1. Software Part:

Train, fix, evaluate, and export your TinyML gesture classification model.

2. Hardware Part:

Deploy the model on the Arduino board and test gesture classification using the Serial Monitor.

- **What to Include in Your Submission PDF**

Create a short **Word document** and include:

- **Screenshots showing both ASL gestures correctly identified** by the Arduino Serial Monitor.
- A brief explanation of the **software issue(s) you fixed** in the notebook.
- Any important changes you made to make the notebook, plots, metrics, or model export consistent.

[Then convert the Word document to PDF.](#)



Submission Instructions

Upload to GitHub

Upload the following files to your GitHub repository:

- Completed `.ipynb` notebook
- Submission PDF
- Arduino `.ino` file
- `model.h`
- Any other related files needed to rerun or verify your work

Canvas Submission

- Submit your **GitHub repository link** on Canvas.
- Make sure the link is accessible to the TAs.

